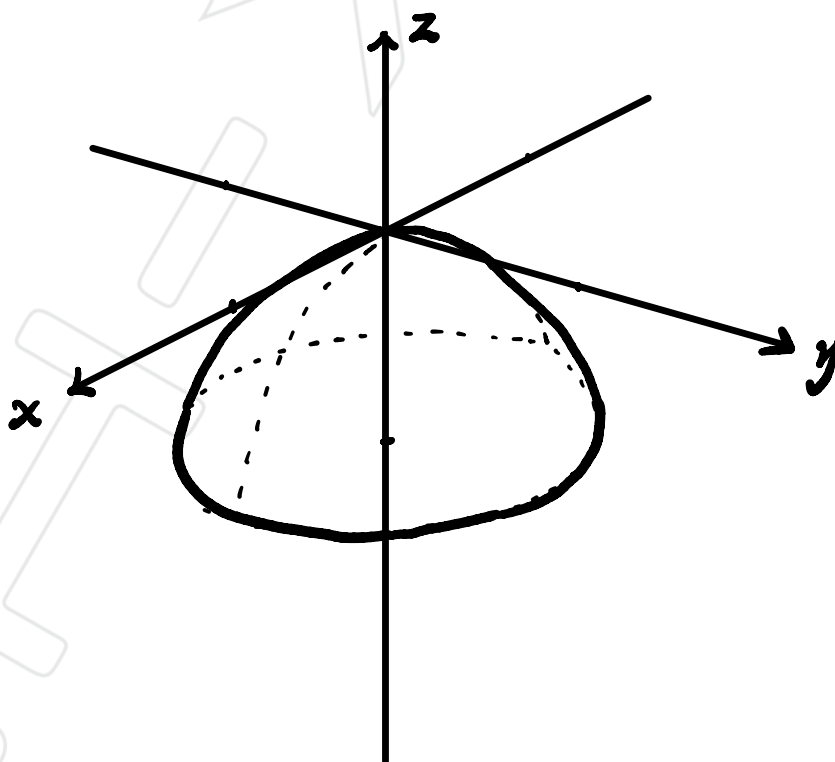
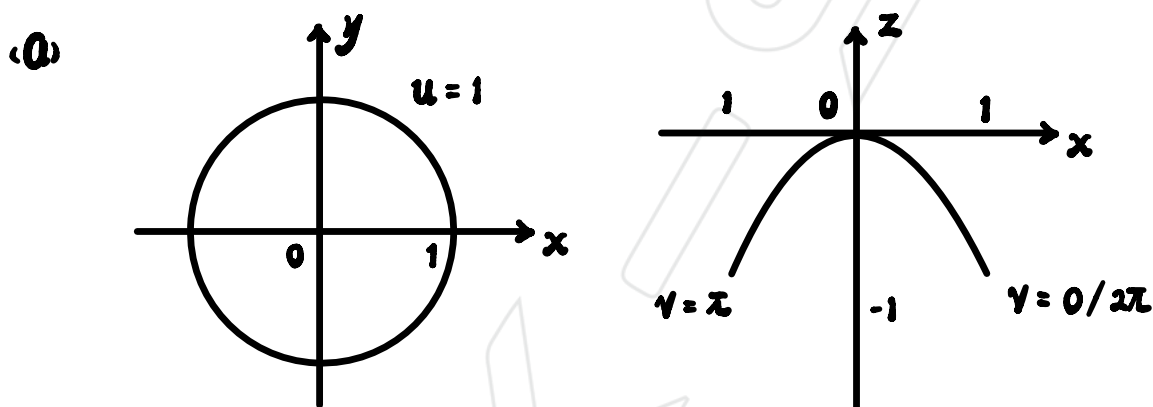


Problem set 4 - Surface integral and integral theorems

1. Consider a parametrised surface S given by $\mathbf{r}(u, v) = (u \cos v, u \sin v, -u^2)$ in $\mathbb{R}^3$ with $0 \leq u \leq 1$ and $0 \leq v \leq 2\pi$.

- Sketch the surface S .
- Find two tangent vectors and the oriented unit normal vector of S .
- Find the area of the surface S .
- Find $\iint_S \mathbf{F} \cdot d\mathbf{S}$ if $\mathbf{F}(x, y, z) = (y, -x, z)$.



$$(b) \quad T_u = (\cos v, \sin v, -2u)$$

$$T_v = (-u \sin v, u \cos v, 0)$$

$$\therefore T_u \times T_v = (2u^2 \cos v, 2u^2 \sin v, u)$$

$$\|T_u \times T_v\| = \sqrt{4u^4 + u^2} = u \sqrt{4u^2 + 1}$$

$$\therefore n = \frac{T_u \times T_v}{\|T_u \times T_v\|} = \left(\frac{2u \cos v}{\sqrt{4u^2 + 1}}, \frac{2u \sin v}{\sqrt{4u^2 + 1}}, \frac{1}{\sqrt{4u^2 + 1}} \right)$$

$$(c) \quad A = \iint_S dA = \iint_A \|T_u \times T_v\| \, du \, dv$$

$$= \int_0^{2\pi} \int_0^1 u \sqrt{4u^2 + 1} \, du \, dv$$

$$\text{Let } a = 4u^2 + 1 \quad \therefore \frac{da}{du} = 8u \quad \therefore du = \frac{da}{8u}$$

$$\therefore A = \int_0^{2\pi} \int_1^5 \frac{1}{8} a^{\frac{1}{2}} da \, dv$$

$$= \int_0^{2\pi} \left[\frac{1}{2} a^{\frac{3}{2}} \right]_1^5 dv$$

$$= \int_0^{2\pi} \left(\frac{5}{12} \sqrt{5} - \frac{1}{12} \right) dv = \frac{5}{6} \sqrt{5} \pi - \frac{1}{6} \pi$$

$$(d) \iint_S \mathbf{F} \cdot d\mathbf{s}$$

$$= \iint_S \mathbf{F} \cdot \mathbf{n} \, dA$$

$$= \iint_S (u \sin v, -u \cos v, -u^2) \cdot \left(\frac{2u \cos v}{\sqrt{4u^2+1}}, \frac{2u \sin v}{\sqrt{4u^2+1}}, \frac{1}{\sqrt{4u^2+1}} \right) dA$$

$$= \iint_S \frac{-u^2}{\sqrt{4u^2+1}} \, dA$$

$$= \int_0^{2\pi} \int_0^1 \frac{-u^2}{\sqrt{4u^2+1}} \cdot u \sqrt{4u^2+1} \, du \, dv$$

$$= \int_0^{2\pi} \int_0^1 -u^3 \, du \, dv$$

$$= \int_0^{2\pi} \left[-\frac{1}{4}u^4 \right]_0^1 \, dv$$

$$= \int_0^{2\pi} -\frac{1}{4} \, dv$$

$$= -\frac{1}{2}\pi$$

2. Consider a parametrised surface S given by $\mathbf{r}(u, v) = (\cos v \cos u, \cos v \sin u, 2 \sin v)$ in $\mathbb{R}^3$ with $0 \leq u \leq 2\pi$ and $-\pi/2 \leq v \leq \pi/2$.

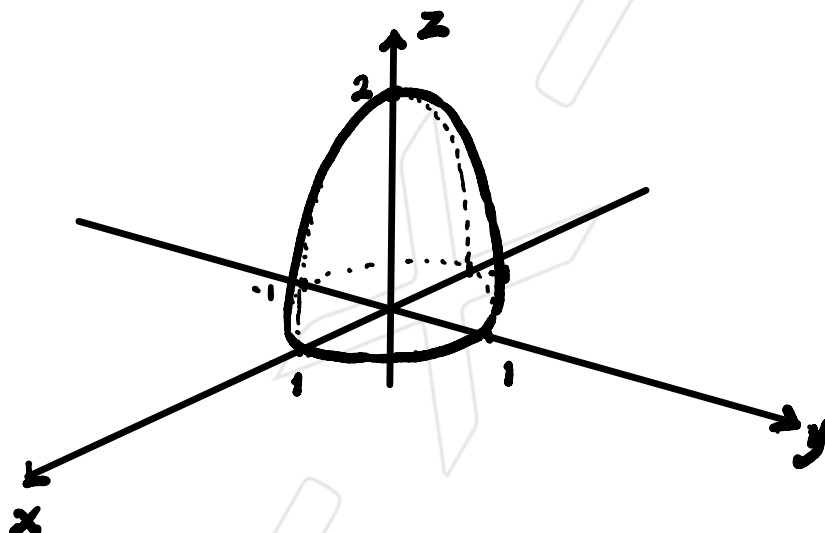
(a) Sketch the surface S .

(b) Find two tangent vectors and the oriented unit normal vector of S .

(c) Find $\iint_S \mathbf{F} \cdot d\mathbf{S}$ if $\mathbf{F}(x, y, z) = (-3y, 3x, 0)$.

$$(a) \quad u = 0, \quad \mathbf{r}(0, v) = (\cos v, 0, 2 \sin v)$$

$$v = 0, \quad \mathbf{r}(u, 0) = (\cos u, \sin u, 0)$$



$$(b) \quad T_u = (-\cos v \sin u, \cos v \cos u, 0)$$

$$T_v = (-\sin v \cos u, -\sin v \sin u, 2 \cos v)$$

$$\therefore T_u \times T_v = (2 \cos^2 v \cos u, 2 \cos^2 v \sin u, \cos v \sin v)$$

$$\begin{aligned}
 \therefore \|T_u \times T_v\| &= (4\cos^4 v + \cos^2 v \sin^2 v)^{\frac{1}{2}} \\
 &= \cos v (4\cos^2 v + \sin^2 v)^{\frac{1}{2}} \\
 &= \cos v (1 + 3\cos^2 v)^{\frac{1}{2}}
 \end{aligned}$$

$$\begin{aligned}
 \therefore n &= \frac{T_u \times T_v}{\|T_u \times T_v\|} \\
 &= \left(\frac{2\cos v \cos u}{(1+3\cos^2 v)^{\frac{1}{2}}}, \frac{2\cos v \sin u}{(1+3\cos^2 v)^{\frac{1}{2}}}, \frac{\sin v}{(1+3\cos^2 v)^{\frac{1}{2}}} \right)
 \end{aligned}$$

$$(c) \iint_S F \cdot ds$$

$$= \iint_S F \cdot n dA = \iint F \cdot n \|T_u \times T_v\| du dv$$

$$= \iint (-3\cos v \sin u, 3\cos v \cos u, 0) \cdot (2\cos v \cos u, 2\cos v \sin u, \cos v \sin v) du dv$$

$$= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \int_0^{2\pi} -6\cos^2 v \sin u \cos u + 6\cos^2 v \sin u \cos u + 0 \, du dv$$

$$= 0$$

$$(\nabla \cdot F = 0, \iint_S F \cdot ds = \iiint_V \nabla \cdot F \, dV = 0)$$

3. Let $z = f(x, y)$ be an open surface S in $\mathbb{R}^3$ and $(x, y) \in D$ in $\mathbb{R}^2$. We parametrise the surface S such that

$$x = u, \quad y = v, \quad z = f(u, v).$$

- (a) Using the parametrisation, find the two tangential vectors on S given by

$$\mathbf{T}_u = \frac{\partial x}{\partial u} \mathbf{i} + \frac{\partial y}{\partial u} \mathbf{j} + \frac{\partial z}{\partial u} \mathbf{k}; \quad \mathbf{T}_v = \frac{\partial x}{\partial v} \mathbf{i} + \frac{\partial y}{\partial v} \mathbf{j} + \frac{\partial z}{\partial v} \mathbf{k}.$$

- (b) Using the answer in (a), show that the surface integral of a scalar-valued function $g(x, y, z)$ is given by

$$\iint_S g(x, y, z) dA = \iint_D g(x, y, f(x, y)) \sqrt{1 + \left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2} dx dy.$$

- (c) Now, let the surface S be given by $z = x^2 + y$ with the domain D defined by $0 \leq x \leq 1$ and $-1 \leq y \leq 1$. Using the formula derived in (b), evaluate $\iint_S x dA$.

$$(a) \quad \mathbf{T}_u = \left(1, 0, \frac{\partial f}{\partial u} \right)$$

$$\mathbf{T}_v = \left(0, 1, \frac{\partial f}{\partial v} \right)$$

$$(b) \quad \iint_S g(x, y, z) dA$$

$$= \iint g(x, y, z) \cdot \|\mathbf{T}_u \times \mathbf{T}_v\| du dv$$

$$= \iint g(x, y, z) \cdot \left\| -\frac{\partial f}{\partial u}, -\frac{\partial f}{\partial v}, 1 \right\| du dv$$

$$\therefore g(x) = u, \quad g(y) = v, \quad g(z) = f(u, v)$$

$$\therefore \iint_S g(x, y, z) \, dA$$

$$= \iint_D g(x, y, z) \sqrt{1 + \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2} \, dx \, dy$$

$$(c) \iint_S x \, dA$$

$$= \iint_D x \sqrt{1 + \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2} \, dx \, dy$$

$$= \int_{-1}^1 \int_0^1 x \sqrt{1 + (2x)^2 + 1} \, dx \, dy$$

$$\text{Let } u = 4x^2 + 2 \quad \therefore \frac{du}{dx} = 8x \quad \therefore dx = \frac{du}{8x}$$

$$\therefore \iint_S x \, dA = \int_{-1}^1 \int_2^6 \frac{1}{8} u^{\frac{1}{2}} \, du \, dy$$

$$= \int_{-1}^1 \left[\frac{1}{12} u^{\frac{3}{2}} \right]_2^6 \, dy$$

$$= \int_{-1}^1 \left[\frac{\sqrt{6}}{2} - \frac{\sqrt{2}}{6} \right] \, dy = \sqrt{6} - \frac{\sqrt{2}}{3}$$

4. Let $z = h(x, y)$ be an open surface S in $\mathbb{R}^3$ and $(x, y) \in D$ in $\mathbb{R}^2$.

- By considering the surface S as a level surface (or a level set), find the oriented unit vector normal to the surface S (Hint: use the gradient of the level surface).
- We consider the parametrisation given in the problem 3: i.e. $x = u$, $y = v$, $z = h(u, v)$. Using this parametrisation, find the oriented unit vector normal to the surface S and compare it with the answer of (a).
- Suppose a vector field $\mathbf{F} : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is given by $\mathbf{F} = (F_1, F_2, F_3)$. Then, using the answer in (b), show

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_D \left[F_1 \left(-\frac{\partial h}{\partial x} \right) + F_2 \left(-\frac{\partial h}{\partial y} \right) + F_3 \right] dx dy.$$

- Now, let $\mathbf{F} = (x^2, 0, 3y^2)$ and the surface S be the portion of the plane $x + y + z = 1$ in the first octant (i.e. $x \geq 0$, $y \geq 0$, and $z \geq 0$). Using the formula in (c), find $\iint_S \mathbf{F} \cdot d\mathbf{S}$.

(a) ~~$f: x + y + z = c$~~ $f: z - h(x, y) = 0$

$$\therefore \mathbf{n} = \frac{\nabla f}{\|\nabla f\|}$$

$$= \left(-\frac{\frac{df}{dx}}{\sqrt{1 + \left(\frac{df}{dx}\right)^2 + \left(\frac{df}{dy}\right)^2}}, -\frac{\frac{df}{dy}}{\sqrt{1 + \left(\frac{df}{dx}\right)^2 + \left(\frac{df}{dy}\right)^2}}, \frac{1}{\sqrt{1 + \left(\frac{df}{dx}\right)^2 + \left(\frac{df}{dy}\right)^2}} \right)$$

(b) for $\mathbf{F} = (x, y, z)$

$$\mathbf{T}_x = (1, 0, \frac{df}{dx}), \quad \mathbf{T}_y = (0, 1, \frac{df}{dy})$$

$$\therefore \mathbf{T}_x \times \mathbf{T}_y = \left(-\frac{df}{dx}, -\frac{df}{dy}, 1 \right)$$

$$\|\mathbf{T}_x \times \mathbf{T}_y\| = \left(1 + \left(\frac{df}{dx}\right)^2 + \left(\frac{df}{dy}\right)^2 \right)^{\frac{1}{2}}$$

$$\therefore n = \frac{T_x \times T_y}{\|T_x \times T_y\|}$$

$$= \left(-\frac{\frac{dz}{dx}}{\sqrt{1 + \left(\frac{dz}{dx}\right)^2 + \left(\frac{dz}{dy}\right)^2}}, -\frac{\frac{dz}{dy}}{\sqrt{1 + \left(\frac{dz}{dx}\right)^2 + \left(\frac{dz}{dy}\right)^2}}, \frac{1}{\sqrt{1 + \left(\frac{dz}{dx}\right)^2 + \left(\frac{dz}{dy}\right)^2}} \right)$$

$$(c) \iint_S F \cdot dS = \iint_S F \cdot n \, dA$$

$$= \iint_D F \cdot (T_x \times T_y) \, dx \, dy$$

$$= \iint_D F_1 \left(-\frac{dz}{dx}\right) + F_2 \left(-\frac{dz}{dy}\right) + F_3 \, dx \, dy$$

$$z = 1 - x - y$$

$$(d) \iint_S F \cdot dS = \int_0^{1-x} \int_0^1 x^2 \cdot 0 + 0 \cdot (-6y) + 3y^2 \, dx \, dy$$

$$= \int_0^1 3y^2 \, dy = \left[y^3 \right]_0^1 = \frac{1}{3}$$

$$= 1$$

5. Let $P(x, y) = x$ and $Q(x, y) = xy$, where x and y are defined on the unit disk D given by $x^2 + y^2 \leq 1$ with the boundary denoted by ∂D .

(a) Evaluate $\int_{\partial D} P dx + Q dy$.

(b) Evaluate $\iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy$ and compare it with the answer in (a).

$$(a) \int_{\partial D} P dx + Q dy = \int_{\partial D} (P, Q) \cdot \frac{d(\partial D)}{dt} dt$$

$$\text{At boundary, } x^2 + y^2 = r^2 = 1$$

$$\therefore \partial D = (\cos t, \sin t) \text{ for } [0, 2\pi]$$

$$\therefore \int_{\partial D} P dx - Q dy$$

$$= \int_0^{2\pi} (\cos t, \cos t \sin t) \cdot (-\sin t, \cos t) dt$$

$$= \int_0^{2\pi} -\sin t \cos t + \cos^2 t \sin t dt$$

$$= \int_0^{2\pi} -\frac{1}{2} \sin(2t) + \frac{1}{2} \sin(2t) \cos t dt$$

$$= 0$$

$$(b) \iint_D \left(\frac{dQ}{dx} - \frac{dP}{dy} \right) dx dy$$

$$= \iint_D y \, dx dy$$

$$= \int_0^{2\pi} \int_0^1 r \sin \theta \cdot r \, dr d\theta$$

$$= \int_0^{2\pi} \int_0^1 r^2 \sin \theta \, dr d\theta$$

$$= \int_0^{2\pi} \left[\frac{1}{3} r^3 \sin \theta \right]_0^1 d\theta$$

$$= \int_0^{2\pi} \frac{1}{3} \sin \theta \, d\theta$$

$$= \left[-\frac{1}{3} \cos \theta \right]_0^{2\pi}$$

$$= -\frac{1}{3} + \frac{1}{3} = 0$$

$$\therefore \int_{\partial S} P dx + Q dy = \iint_S \left(\frac{dQ}{dx} - \frac{dP}{dy} \right) dx dy$$

6. Compute the area of the region enclosed by $x^{2/3} + y^{2/3} = a^{2/3}$ using the parametrisation

$$x = a \cos^3 \theta, \quad y = a \sin^3 \theta, \quad (1)$$

for $0 \leq \theta \leq 2\pi$. (Hint: use the Green's theorem).

$$A = \frac{1}{2} \int_0^{2\pi} r^2 d\theta$$

$$x^{2/3} + y^{2/3} = a^{2/3} = r^2 \quad \therefore r = a^{1/3}$$

$$\therefore x' = x^{1/3} = a^{1/3} \cos \theta, \quad y' = y^{1/3} = a^{1/3} \sin \theta$$

$$\therefore A = \frac{1}{2} \int_0^{2\pi} r^2 d\theta$$

$$= \frac{1}{2} \int_0^{2\pi} a^{2/3} d\theta$$

$$= a^{2/3} \pi$$

6. Compute the area of the region enclosed by $x^{2/3} + y^{2/3} = a^{2/3}$ using the parametrisation

$$x = a \cos^3 \theta, \quad y = a \sin^3 \theta, \quad (1)$$

for $0 \leq \theta \leq 2\pi$. (Hint: use the Green's theorem).

$$A = \frac{1}{2} \int x dy - y dx$$

$$= \frac{1}{2} \int \left(x \frac{dy}{d\theta} - y \frac{dx}{d\theta} \right) d\theta$$

$$= \frac{1}{2} \int \left((a \cos^3 \theta)(3a \cos \theta \sin^2 \theta) - (a \sin^3 \theta)(-3a \sin \theta \cos^2 \theta) \right) d\theta$$

$$= \frac{3}{2} a^2 \int_0^{2\pi} (\cos^4 \theta \sin^2 \theta + \sin^4 \theta \cos^2 \theta) d\theta$$

$$= \frac{3}{2} a^2 \int_0^{2\pi} \cos^2 \theta \sin^2 \theta (\cos^2 \theta + \sin^2 \theta) d\theta$$

$$= \frac{3}{2} a^2 \int_0^{2\pi} \cos^2 \theta \sin^2 \theta d\theta$$

$$= \frac{3}{2} a^2 \cdot \frac{\pi}{4}$$

$$= \frac{3}{8} \pi a^2$$

7. Let S be the part of the cylinder $z = 1 - x^2$ for $0 \leq x \leq 1$, $-2 \leq y \leq 2$, and the boundary be denoted by ∂S . Also, suppose a vector field given by $\mathbf{F} = (xy, yz, xz)$.

(a) Evaluate $\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}$.

(b) Evaluate $\int_{\partial S} \mathbf{F} \cdot d\mathbf{s}$ and compare it with the answer of (a).

$$(a) \nabla \times \mathbf{F}$$

$$= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xy & yz & xz \end{vmatrix}$$

$$= (-y, -z, -x)$$

$$\therefore \iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}$$

$$= \iint_S F_1 \left(-\frac{\partial}{\partial x}\right) + F_2 \left(-\frac{\partial}{\partial y}\right) + F_3 \, dx dy$$

$$= \int_{-2}^2 \int_0^1 (-y) \cdot (2x^2) + (-z) \cdot 0 - x \, dx dy$$

$$= \int_{-2}^2 \int_0^1 -2x^2y - x \, dx dy = \int_{-2}^2 -\frac{2}{3}y - \frac{1}{2} \, dy$$

$$= \left[-\frac{1}{3}y^2 - \frac{1}{2}y\right]_{-2}^2 = -\frac{7}{3} - \left(-\frac{1}{3}\right) = -2$$

$$(b) \int_{\partial S} \mathbf{F} \cdot d\mathbf{s} = \int_{\partial S} (xy, yz, xz) \cdot \frac{d(\mathbf{s})}{dt} dt$$

① At boundary condition, $z=0$, $x^2=1$

$$\therefore d\mathbf{s} = (\cancel{1}, \cancel{t}, 0)$$

$$\therefore \int_{\partial S} \mathbf{F} \cdot d\mathbf{s} = 0$$

② At boundary condition, $x=0$, $z=1$

$$\therefore d\mathbf{s} = (0, t, 1)$$

$$\therefore \int_{\partial S} \mathbf{F} \cdot d\mathbf{s} = 0$$

③ At boundary condition, $y=2$, $z=1-x^2$

$$\therefore d\mathbf{s} = (t, 2, 1-t^2)$$

$$\therefore \int_{\partial S} \mathbf{F} \cdot d\mathbf{s} = \int_1^0 2t + 0 + (t-t^3)(-2t) dt = -\frac{11}{15}$$

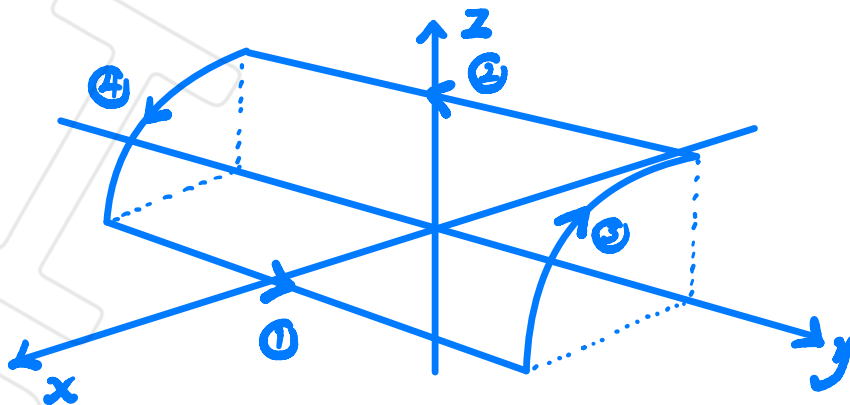
④ At boundary condition, $y = -2$, $z = 1 - x^2$

$$\therefore d\mathbf{s} = (t, -2, 1-t^2)$$

$$\therefore \int_{\partial S} \mathbf{F} \cdot d\mathbf{s} = \int_0^1 -2t + 0 + (t-t^2)(-2t) dt = -\frac{14}{15}$$

$$\therefore \int_{\partial S} \mathbf{F} \cdot d\mathbf{s} = 0 + 0 - \frac{11}{15} - \frac{14}{15} = -2$$

$$\therefore \iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{s} = \int_{\partial S} \mathbf{F} \cdot d\mathbf{s}$$



8. Let $\mathbf{F}(x, y, z) = (xy^2, x^2y, y)$ and S be the surface of the cylinder $x^2 + y^2 = 1$, bounded by the planes $z = 1$ and $z = -1$ and including the portions $x^2 + y^2 \leq 1$ when $z = \pm 1$. Evaluate $\iint_S \mathbf{F} \cdot d\mathbf{S}$ using the Gauss divergence theorem.

$$\begin{aligned}
 \iint_S \mathbf{F} \cdot d\mathbf{S} &= \iiint_V \nabla \cdot \mathbf{F} \, dV \\
 &= \iiint_V y^2 + x^2 \, dV \\
 &= \int_{-1}^1 \int_0^{2\pi} \int_0^1 [(\cos\theta)^2 + (\sin\theta)^2] r \, dr \, d\theta \, dz \\
 &= \int_{-1}^1 \int_0^{2\pi} \int_0^1 r^2 \, dr \, d\theta \, dz \\
 &= \int_{-1}^1 \int_0^{2\pi} \frac{1}{3} \, d\theta \, dz \\
 &= \int_{-1}^1 \frac{1}{3} \cdot 2\pi \, dz \\
 &= \pi
 \end{aligned}$$

9. Suppose a temperature function is given as $T(x, y, z) = x^2 + y^2 + z^2$, and let S be the unit sphere $x^2 + y^2 + z^2 = 1$. The heat flux across the surface S is given by $\mathbf{F} = -k\nabla T$ (k is thermal conductivity).

- (a) Calculate the total heat flux across the surface S (Hint: calculate $\iint_S \mathbf{F} \cdot d\mathbf{S}$).
 (b) Now, calculate the total heat flux again using the Gauss divergence theorem, and compare it with the answer of (a).

$$(a) \iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_S -k\nabla T \cdot d\mathbf{S}$$

$$= -k \iint_S (2x, 2y, 2z) \cdot d\mathbf{S}$$

$$\therefore x^2 + y^2 + z^2 = r^2 = 1$$

$$\therefore \mathbf{n} = \frac{\nabla f}{\|\nabla f\|} = \frac{(2x, 2y, 2z)}{2} = (x, y, z)$$

$$\therefore \iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_S \mathbf{F} \cdot \mathbf{n} \, dA$$

$$= -k \iint_S 2x^2 + 2y^2 + 2z^2 \, dA$$

$$= -k \iint_S 2 \, dA = -2k \cdot A$$

$$= -2k(4\pi)$$

$$= -8k\pi$$

$$(b) \iint_S \mathbf{F} \cdot d\mathbf{s} = \iiint_V \nabla \cdot \mathbf{F} \, dV$$

$$= -k \iiint_V \nabla \cdot (2x, 2y, 2z) \, dV$$

$$= -k \iiint_V (2+2+2) \, dV$$

$$= -6k \cdot V = -6k \cdot \left(\frac{4}{3}\pi\right)$$

$$= -8k\pi$$